Damn Fast DFIR — Router & Firewall Cheat Sheet

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Vyatta / VyOS first, then Cisco IOS and ASA, Junos, FortiGate, PAN-OS, MikroTik, pfSense, Linux and Windows: safe rule changes, blocking, containment, verification and on-box DFIR.

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src: src/routers_content.html

QUICK START Ban an IP address or range, step by step (Vyatta / VyOS)

Example: ban **198.51.100.7**, the network **203.0.113.0/24** and the range **192.0.2.10-192.0.2.20** arriving on the internet interface. VyOS 1.4+: write `firewall ipv4 name` instead of `firewall name`.

1. **Find the rule-sets on the internet interface** (op mode, prompt `$`). Pick free rule numbers below the first allow rule (here 5-7).

```
show configuration commands | match firewall   # e.g. eth0
   firewall in name WAN_IN / local name WAN_LOCAL
show firewall name WAN_IN                      # existing rule
   numbers
```

2. **Enter config mode and take a snapshot**

```
configure
save /config/pre-ban-CR123.boot
```

3. **Ban one IP:** `WAN_IN` covers traffic through the router, and `WAN_LOCAL` covers traffic to the router itself.

```
set firewall name WAN_IN rule 5 action drop
set firewall name WAN_IN rule 5 source address 198.51.100.7
set firewall name WAN_IN rule 5 log enable
set firewall name WAN_IN rule 5 description 'BAN 198.51.100.7
   CR123'
set firewall name WAN_LOCAL rule 5 action drop
set firewall name WAN_LOCAL rule 5 source address 198.51.100.7
```

4. **Ban a range:** CIDR network (rule 6) and first-last range (rule 7). Repeat the lines for `WAN_LOCAL`.

```
set firewall name WAN_IN rule 6 action drop
set firewall name WAN_IN rule 6 source address 203.0.113.0/24
set firewall name WAN_IN rule 6 log enable
set firewall name WAN_IN rule 6 description 'BAN 203.0.113.0/24
   CR123'
set firewall name WAN_IN rule 7 action drop
set firewall name WAN_IN rule 7 source address 192.0.2.10-
   192.0.2.20
set firewall name WAN_IN rule 7 log enable
set firewall name WAN_IN rule 7 description 'BAN 192.0.2.10-20
   CR123'
```

5. **Optional: stop inside hosts reaching them** (egress, LAN rule-set)

```
set firewall name LAN_IN rule 5 action drop
set firewall name LAN_IN rule 5 destination address
   198.51.100.7
set firewall name LAN_IN rule 5 log enable
```

6. **Review.** Only the new `+` lines should appear.

```
compare
```

7. **Apply with a safety net.** The change auto-reverts in 10 minutes unless you confirm it.

```
commit-confirm 10
```

8. **Verify:** the counters on rules 5-7 rise and the log shows drops.

```
run show firewall name WAN_IN statistics
run show log | match WAN_IN-5
```

9. **Keep it** and go back to op mode.

```
confirm
save
exit
```

10. **Cut sessions that were already open.** The ban only stops new ones.

```
sudo conntrack -D -s 198.51.100.7
sudo conntrack -D -s 203.0.113.0 --mask-src 255.255.255.0
for i in $(seq 10 20); do sudo conntrack -D -s 192.0.2.$i; done
```

11. **Unban later**

```
configure
delete firewall name WAN_IN rule 5
delete firewall name WAN_LOCAL rule 5
commit
save
exit
```

Many bans? Point one drop rule at a group, then each new ban is one line: `set firewall group address-group BLOCKLIST address 198.51.100.7` (IPs and first-last ranges) or `set firewall group network-group BLOCKNETS network 203.0.113.0/24` (CIDR). See the IR recipes card.

QUICK START The same ban on other kit, step by step

Same examples: **198.51.100.7**, **203.0.113.0/24**, **192.0.2.10-192.0.2.20**. Back up the config first (see Rosetta › Back up).

CISCO IOS / IOS-XE (ACL ALREADY ON THE OUTSIDE INTERFACE)

```
configure terminal
ip access-list extended WAN_IN
 5 deny ip host 198.51.100.7 any log
 6 deny ip 203.0.113.0 0.0.0.255 any log          ! wildcard mask,
   not netmask
end
show access-lists WAN_IN                          ! check order +
   matches
write memory
```

IOS ACLs have no first-last ranges. Use CIDR blocks, or on IOS-XE `object-group network BANNED` / `range 192.0.2.10 192.0.2.20` with `deny ip object-group BANNED any log`.

CISCO ASA

```
configure terminal
object network BAN-RANGE1
 range 192.0.2.10 192.0.2.20
object-group network BANNED
 network-object host 198.51.100.7
 network-object 203.0.113.0 255.255.255.0
 network-object object BAN-RANGE1
 exit
access-list OUTSIDE_IN line 1 extended deny ip object-group BANNED
   any log
end
clear conn address 198.51.100.7                   ! cut live
   sessions
write memory
```

FORTIGATE

```
config firewall address
    edit "BAN-IP1"
        set subnet 198.51.100.7 255.255.255.255
    next
    edit "BAN-NET1"
        set subnet 203.0.113.0 255.255.255.0
    next
    edit "BAN-RANGE1"
        set type iprange
        set start-ip 192.0.2.10
        set end-ip 192.0.2.20
    next
end
config firewall addrgrp
    edit "BANNED"
        set member "BAN-IP1" "BAN-NET1" "BAN-RANGE1"
    next
end
config firewall policy
    edit 0
        set name "BAN-inbound"
        set srcintf "wan1"
        set dstintf "any"
        set srcaddr "BANNED"
        set dstaddr "all"
        set action deny
        set schedule "always"
        set service "ALL"
        set logtraffic all
    next
end
show firewall policy                    # note the new ID, then:
   config firewall policy / move <new> before <top> / end
```

MIKROTIK ROUTEROS

```
/ip firewall address-list add list=BANNED address=198.51.100.7
/ip firewall address-list add list=BANNED address=203.0.113.0/24
/ip firewall address-list add list=BANNED address=192.0.2.10-
   192.0.2.20
/ip firewall raw add chain=prerouting src-address-list=BANNED
   action=drop place-before=0
/ip firewall raw print stats            # raw drops before
   conntrack, so live sessions die too
```

LINUX (IPTABLES)

```
iptables -I INPUT 1 -s 198.51.100.7 -j DROP
iptables -I INPUT 1 -s 203.0.113.0/24 -j DROP
iptables -I INPUT 1 -m iprange --src-range 192.0.2.10-192.0.2.20 -j
   DROP
# routers/gateways: repeat with FORWARD instead of INPUT
iptables -L INPUT -n -v --line-numbers
```

QUICK START The same ban on other kit, step by step (cont.)

```
conntrack -D -s 198.51.100.7
iptables-save > /etc/iptables/rules.v4       # persist
   (Debian/Ubuntu)
```

WINDOWS HOST FIREWALL

```
New-NetFirewallRule -DisplayName "BAN CR123 in"  -Direction Inbound
   -Action Block `
  -RemoteAddress 198.51.100.7,203.0.113.0/24,192.0.2.10-192.0.2.20
New-NetFirewallRule -DisplayName "BAN CR123 out" -Direction Outbound
   -Action Block `
  -RemoteAddress 198.51.100.7,203.0.113.0/24,192.0.2.10-192.0.2.20
Get-NetFirewallRule -DisplayName "BAN CR123*"     # verify; |
   Remove-NetFirewallRule to unban
```

Junos, PAN-OS and pfSense: see their vendor cards and Rosetta › Block one IP. Ranges there are address objects (Junos `address-range`, PAN-OS `ip-range`, pfSense alias).

VYATTA / VYOS Modes & the commit model

```
vyos@fw:~$ show version              # op mode ($): know your
   dialect first
vyos@fw:~$ configure                 # into config mode (#)
vyos@fw# edit firewall name WAN_IN   # move down the tree; up / top
   to go back
vyos@fw# set ...   delete ...       # stage changes in the WORKING
   config
vyos@fw# compare                     # diff working vs running
vyos@fw# commit-confirm 10           # apply; auto-revert in 10 min
   ...
vyos@fw# confirm                     # ... unless confirmed
vyos@fw# commit                      # apply without the safety net
vyos@fw# save                        # write /config/config.boot
   (survives reboot)
vyos@fw# discard   /   exit discard   # throw away uncommitted
   changes
vyos@fw# run show firewall           # op command from config mode
```

COMMAND	DOES
`?` `Tab`	list options / complete
`show configuration commands | match firewall`	(op) running config as set-commands, filtered
`... | strip-private  | no-more`	hide secrets (VyOS 1.3+) / no pager
`show`	(config) the subtree at the current edit level
`copy rule 10 to rule 11`	clone a rule (inside `edit firewall name X`)
`rename rule 10 to rule 15`	renumber, i.e. reorder (first match wins)
`comment firewall name WAN_IN rule 10 "CR123"`	config comment
`save /config/pre-CR123.boot  load <file>  merge <file>`	snapshot / replace working config / merge into it
`show system commit  compare 0 3  rollback 3`	(op) revisions / (config) diff two revisions / revert (needs reboot)

commit-confirm reverts by rebooting into the *saved* config.boot (Vyatta, EdgeOS, VyOS ≤1.3). Do not `save` until you have confirmed. Unsaved earlier commits are lost on revert as well.

VYATTA / VYOS Change procedure (firewall rule change)

1. **Prepare:** change ref, rollback plan, console / out-of-band access. Know which session could be cut.
2. **Snapshot:** `configure` then `save /config/pre-CR123.boot`. Also keep `run show configuration commands` in the CR.
3. **Baseline:** `run show firewall name WAN_IN statistics` (counters), `run show interfaces`.
4. **Stage:** enter the `set` / `delete` lines. Leave number gaps (10, 20, 30) so rules can be inserted later.
5. **Review:** `compare`. Check rule order, state rules, and the default-action.
6. **Apply with a timer:** `commit-confirm 10`.
7. **Verify:** test new sessions from allowed and denied sources. Watch counters and `run show log | match WAN_IN`. Check that your management session still works.
8. **Keep it:** `confirm`, then `save`. Record `run show system commit` and the `compare` output in the CR.
9. **Back out:** `load /config/pre-CR123.boot`, `commit`, `save` (no reboot needed).

VYATTA / VYOS Firewall model (Vyatta 6.x, EdgeOS, VyOS ≤1.3)

- A **rule-set** (`firewall name X`, IPv6 `ipv6-name X`) holds numbered rules (1-9999). Rules are evaluated in ascending order and the **first match wins**. Unmatched traffic gets the `default-action`, which is **drop** unless set.

VYATTA / VYOS Firewall model (Vyatta 6.x, EdgeOS, VyOS ≤1.3) (cont.)

- Bind at most one rule-set per interface per direction: **in** = forwarded traffic arriving on the interface, **out** = forwarded traffic leaving it, **local** = traffic to the router itself (SSH, DNS, GUI, VPN).
- The firewall is stateful only if you allow it: add established/related accept (per rule-set, or the global `state-policy` on VyOS 1.2+). Without it, reply traffic is dropped.
- Destination NAT happens **before** `in` filtering, so WAN_IN rules match the **internal** (translated) address.

```
# skeleton: stateful default-deny WAN
set firewall name WAN_IN default-action drop
set firewall name WAN_IN rule 10 action accept
set firewall name WAN_IN rule 10 state established enable
set firewall name WAN_IN rule 10 state related enable
set firewall name WAN_IN rule 20 action drop
set firewall name WAN_IN rule 20 state invalid enable
set firewall name WAN_LOCAL default-action drop
set firewall name WAN_LOCAL rule 10 action accept
set firewall name WAN_LOCAL rule 10 state established enable
set firewall name WAN_LOCAL rule 10 state related enable
set interfaces ethernet eth0 firewall in name WAN_IN
set interfaces ethernet eth0 firewall local name WAN_LOCAL
set interfaces ethernet eth1 vif 20 firewall in name GUEST_IN   #
   VLAN sub-interface
```

VYATTA / VYOS Rule options

All under `set firewall name WAN_IN rule 30 ...`

OPTION	MEANING
`action accept | drop | reject`	reject sends RST / ICMP unreachable
`description 'CR123 allow HTTPS to web1'`	quote values containing spaces
`protocol tcp | udp | tcp_udp | icmp | all | 47`	name or number
`source address 203.0.113.5 | 10.0.0.0/24 | 10.0.0.1-10.0.0.9 | !10.0.0.0/8`	host / net / range / negated
`destination port 22 | 80,443 | 8000-8100 | https`	needs `protocol tcp|udp|tcp_udp`
`source group address-group BLOCKLIST`	also `network-group`, `port-group`
`state established | related | new | invalid enable`	conntrack state match
`icmp type-name echo-request  (icmp type 8)`	ICMP type (VyOS / classic Vyatta)
`tcp flags 'SYN,!ACK'`	flag match (VyOS 1.2/1.3)
`recent count 4  recent time 60`	4+ hits in 60 s (brute-force throttle)
`limit rate 10/minute  limit burst 5`	rate limit
`time starttime 08:00:00 stoptime 18:00:00 weekdays Mon,Tue`	time window
`source mac-address 00:11:22:33:44:55`	MAC match
`log enable  disable`	log hits / switch a rule off without deleting it
`set firewall name X enable-default-log`	log what the default-action drops

```
# groups: change members without touching rules
set firewall group address-group BLOCKLIST address 198.51.100.7
set firewall group address-group BLOCKLIST address 198.51.100.10-
   198.51.100.20
set firewall group network-group RFC1918 network 10.0.0.0/8
set firewall group port-group WEB port 80
set firewall group port-group WEB port 443
delete firewall group address-group BLOCKLIST address 198.51.100.7
```

VYATTA / VYOS Global options, zones, NAT

```
# global hardening + state policy (VyOS 1.2/1.3)
set firewall state-policy established action accept
set firewall state-policy related action accept
set firewall state-policy invalid action drop
set firewall all-ping enable
set firewall broadcast-ping disable
set firewall syn-cookies enable
set firewall source-validation strict        # reverse-path / anti-
   spoof
set firewall log-martians enable
set firewall send-redirects disable
set firewall receive-redirects disable
set firewall ip-src-route disable
# zone-based (do not mix with interface rule-sets on the same
   interface)
set zone-policy zone WAN interface eth0
set zone-policy zone LAN interface eth1
set zone-policy zone LAN default-action drop
```

VYATTA / VYOS Global options, zones, NAT (cont.)

```
set zone-policy zone LAN from WAN firewall name WAN-LAN
set zone-policy zone LOCAL local-zone
set zone-policy zone LOCAL from LAN firewall name LAN-LOCAL
# source NAT (masquerade LAN out of eth0)
set nat source rule 100 outbound-interface eth0
set nat source rule 100 source address 10.0.0.0/24
set nat source rule 100 translation address masquerade
# port forward 443 to 10.0.0.10, plus the firewall allow (post-DNAT
   address!)
set nat destination rule 10 inbound-interface eth0
set nat destination rule 10 protocol tcp
set nat destination rule 10 destination port 443
set nat destination rule 10 translation address 10.0.0.10
set firewall name WAN_IN rule 40 action accept
set firewall name WAN_IN rule 40 protocol tcp
set firewall name WAN_IN rule 40 destination address 10.0.0.10
set firewall name WAN_IN rule 40 destination port 443
```

EdgeOS / old Vyatta NAT: `set service nat rule 5010 type masquerade`, `type destination`, `outbound-interface` / `inbound-interface`, `inside-address address`.

VYATTA / VYOS IR recipes: block, contain, lock down

```
# 1. pre-wired IOC blocklist: afterwards a new IOC is ONE line +
   commit
set firewall group address-group BLOCKLIST description 'IR
   blocklist'
set firewall group address-group BLOCKLIST address 198.51.100.7
set firewall name WAN_IN rule 5 action drop
set firewall name WAN_IN rule 5 source group address-group BLOCKLIST
set firewall name WAN_IN rule 5 log enable
set firewall name WAN_LOCAL rule 5 action drop
set firewall name WAN_LOCAL rule 5 source group address-group
   BLOCKLIST
set firewall name LAN_IN rule 5 action drop
set firewall name LAN_IN rule 5 destination group address-group
   BLOCKLIST
set firewall name LAN_IN rule 5 log enable
#    a global state-policy accepts live flows before rule 5, so kill
   them too:
sudo conntrack -D -s 198.51.100.7   ;   sudo conntrack -D -d
   198.51.100.7
# 2. contain host 10.0.0.66: allow only the IR box 10.0.0.5
set firewall name LAN_IN rule 1 action accept
set firewall name LAN_IN rule 1 source address 10.0.0.66
set firewall name LAN_IN rule 1 destination address 10.0.0.5
set firewall name LAN_IN rule 2 action drop
set firewall name LAN_IN rule 2 source address 10.0.0.66
set firewall name LAN_IN rule 2 log enable
set firewall name LAN_LOCAL rule 2 action drop
set firewall name LAN_LOCAL rule 2 source address 10.0.0.66
# 3. SSH to the router from the management subnet only (use commit-
   confirm!)
set firewall group network-group MGMT network 10.0.99.0/24
set firewall name LAN_LOCAL rule 20 action accept
set firewall name LAN_LOCAL rule 20 protocol tcp
set firewall name LAN_LOCAL rule 20 destination port 22
set firewall name LAN_LOCAL rule 20 source group network-group MGMT
set firewall name LAN_LOCAL rule 21 action drop
set firewall name LAN_LOCAL rule 21 protocol tcp
set firewall name LAN_LOCAL rule 21 destination port 22
# 4. throttle SSH brute force on WAN
set firewall name WAN_LOCAL rule 30 action drop
set firewall name WAN_LOCAL rule 30 protocol tcp
set firewall name WAN_LOCAL rule 30 destination port 22
set firewall name WAN_LOCAL rule 30 state new enable
set firewall name WAN_LOCAL rule 30 recent count 4
set firewall name WAN_LOCAL rule 30 recent time 60
```

Containment limit: the router only sees *routed* traffic. Hosts in the same subnet/VLAN talk directly. Contain at the switch (port shutdown / VLAN move / ACL) or via EDR as well.

VYATTA / VYOS Verify & troubleshoot (op mode)

COMMAND	SHOWS
`show firewall`	rule-sets, bindings, counters
`show firewall name WAN_IN [rule 10]`	one rule-set / rule
`show firewall name WAN_IN statistics`	packets/bytes per rule
`clear firewall name WAN_IN counters`	zero counters before a test
`show firewall group`	group members
`show log | match WAN_IN`	logged hits; prefix `[WAN_IN-10-D]` = set-rule-action (A accept, D drop, R reject; `default` = default-action)
`monitor log`	follow the log live
`show conntrack table ipv4 | match 10.0.0.66`	live sessions
`show nat source rules` `show nat destination rules`	NAT rules + counters

VYATTA / VYOS Verify & troubleshoot (op mode) (cont.)

COMMAND	SHOWS
`show interfaces` `show ip route` `show arp`	L2/L3 state
`ping 10.0.0.10 count 3` `traceroute 10.0.0.10`	reachability
`sudo tcpdump -ni eth0 host 198.51.100.7 -w /tmp/c42.pcap`	capture on the box
`sudo iptables -L -n -v --line-numbers`	kernel view (VyOS chains: `NAME_WAN_IN`)

VYATTA FAMILY Dialects: what changes

ITEM	VYATTA 6 / EDGEOS / VYOS ≤1.3	VYOS 1.4+ (NFTABLES)
Rule-set	`firewall name X`	`firewall ipv4 name X`
Bind (forward)	`interfaces ethernet eth0 firewall in name X`	`firewall ipv4 forward filter rule 10 action jump; ... jump-target X; ... inbound-interface name eth0`
Bind (router)	`... firewall local name X`	`firewall ipv4 input filter rule 10 ...` (same jump)
State rule	`state established enable`	`state established`
Global state	`firewall state-policy ...`	`firewall global-options state-policy ...`
Zones	`zone-policy zone LAN ...`	`firewall zone LAN ...`
NAT iface	`outbound-interface eth0`	`outbound-interface name eth0`
Kernel view	`sudo iptables -L -n -v`	`sudo nft list ruleset`

Early 1.4 rolling builds used `inbound-interface interface-name`. **Vyatta 5600 / DANOS** (dataplane): `security firewall name X rule N ...`, bound with `interfaces dataplane dp0p1s1 firewall in X`, groups under `resources group address-group`, NAT under `service nat`. Confirm with `?` on the box.

VYATTA / VYOS DFIR on the router itself

```
show version ; show system uptime ; show users          # who is on
   now
show configuration commands | strip-private > /tmp/cfg.txt
show system commit                                      # revision,
   user, time, via
ls -l /config/archive/                                  #
   config.boot.N.gz per commit
zcat /config/archive/config.boot.3.gz > /tmp/r3.boot    # old
   revision (configure; load /tmp/r3.boot)
show configuration commands | match login               # users +
   public-keys
show configuration commands | match task-scheduler      # scheduled
   persistence
ls -la /config/scripts/ /config/scripts/post-config.d/  # boot
   scripts = persistence
cat /config/scripts/vyos-postconfig-bootup.script       # (vyatta-
   postconfig-bootup.script)
ls -la /home/*/.ssh/ /root/.ssh/ /config/auth/
show log tail 200 ; sudo grep -i sshd /var/log/auth.log
   /var/log/messages
show conntrack table ipv4 ; show arp ; show ip route
```

ANY PLATFORM Rule-change procedure

1. Write the change: source, destination, port, direction, reason, CR ref, owner, expiry date.
2. Check the **NAT position** (table below) and the **rule order**: where does the new rule land, and what matches before it?
3. Get console / out-of-band access. Never rely on the path you are changing.
4. Back up the config to a file, and copy it off the box when possible.
5. Stage, then diff (`compare`, `show | compare`, GUI preview).
6. Apply with an **auto-revert** where the platform has one.
7. Test both ways: allowed traffic passes, denied traffic is blocked and logged, management access survives. Check the hit counters.
8. Confirm, save, record (diff, counters, time), and close the CR.

ANY PLATFORM Rule design rules of thumb

- First match wins on most platforms. Put specific rules before general ones, and blocks before broad allows. (Windows Firewall is the exception: block beats allow.)
- Put the anti-lockout / management allow at the top, and the established/related allow early.
- End with an explicit **deny-all + log**, so dropped traffic is visible. Implicit denies usually have no counter.

ANY PLATFORM Rule design rules of thumb (cont.)

- Use **object groups** for IOC blocks. The rule stays fixed and incident changes only touch membership.
- A new block does **not** kill existing sessions on stateful firewalls. Clear the sessions as well.
- Number in gaps (10, 20, 30). Give every rule a description with the CR ref and date.
- Filter egress as well as ingress. C2 and exfil leave through allowed outbound rules.

ANY PLATFORM Platform models & NAT position

PLATFORM	EVALUATION / DEFAULT	RULE SEES, AFTER DNAT
Vyatta / VyOS	numbered, first match; default-action (drop)	internal (post-DNAT) IP
Cisco IOS ACL	sequence order; implicit `deny any`; stateless	outside-in: public (pre-NAT) IP
Cisco ASA 8.3+	ACL line order; implicit deny; stateful	real (internal) IP
Juniper SRX	zone-pair policies top-down; default deny	internal (post-DNAT) IP
FortiGate	policy list order (not ID); implicit deny	VIP object as dstaddr
Palo Alto	top-down; intrazone allow, interzone deny	pre-NAT IP, post-NAT zone
MikroTik	chain order; **default ACCEPT**	internal (dstnat first)
pfSense / OPNsense	per-interface inbound, first match; WAN default deny	internal (linked NAT rule)
iptables / nft	chain order; chain policy	FORWARD: internal
Windows FW	no order: block > allow; inbound default block	n/a (host firewall)

ROSETTA Auto-revert safety nets

PLATFORM	ARM	KEEP / REVERT
Vyatta / VyOS	`commit-confirm 10`	`confirm`; else reboot to saved config
Junos	`commit confirmed 10`	`commit`; else auto `rollback 1`
Cisco IOS-XE	`configure terminal revert timer 10`	`configure confirm` (needs `archive path`)
Cisco IOS / ASA	`reload in 10`	`reload cancel`; do not `write mem` first
FortiOS	`config system global set cfg-save revert set cfg-revert-timeout 600`	`execute cfg save`; else reverts
MikroTik	`Ctrl+X (Safe Mode)`	Ctrl+X again keeps; a dropped session reverts
Linux iptables	`iptables-apply -t 60 rules.v4`	answer y, or it restores
Linux nft	`(sleep 300; nft -f /root/pre.nft) &`	`kill %1` to keep (pre.nft starts with `flush ruleset`)
PAN-OS / pfSense / Win	none built in	saved config + console; pfSense console option 15

ROSETTA Back up the config

COMMAND	PLATFORM
`save /config/pre-CR123.boot`	Vyatta / VyOS (config mode)
`copy running-config flash:pre-CR123.cfg`	Cisco IOS (`write memory` = save)
`copy running-config disk0:/pre-CR123.cfg`	Cisco ASA
`request system configuration rescue save`	Junos (plus 50 rollbacks kept)
`execute backup config usb pre-CR123.conf`	FortiOS (or GUI › Backup)
`save config to pre-CR123.xml`	PAN-OS (config mode)
`/export file=pre-CR123  /system backup save name=pre-CR123`	MikroTik (text / binary)
`Diagnostics › Backup & Restore`	pfSense / OPNsense (config.xml)
`iptables-save > pre.rules  nft list ruleset > pre.nft`	Linux
`netsh advfirewall export C:\IR\pre.wfw`	Windows

ROSETTA Block one IP fast (198.51.100.7)

COMMAND	PLATFORM
`set firewall group address-group BLOCKLIST address 198.51.100.7`	Vyatta (pre-wired drop rules)
`ip access-list extended WAN_IN  5 deny ip host 198.51.100.7 any log`	Cisco IOS
`shun 198.51.100.7`	Cisco ASA: instant, kills connections, lost on reload
`set security address-book global address-set BLOCKLIST address BAD7`	Junos (address BAD7 defined first)
`config firewall addrgrp / edit BLOCKLIST / append member BAD7`	FortiOS
`set address-group BLOCKLIST static [ BAD7 ]`	PAN-OS (then commit)
`/ip firewall address-list add list=BLOCKLIST address=198.51.100.7`	MikroTik
`pfctl -t BLOCKLIST -T add 198.51.100.7`	pfSense: runtime only, add to the alias to keep
`iptables -I INPUT -s 198.51.100.7 -j DROP` `iptables -I FORWARD -s 198.51.100.7 -j DROP`	Linux iptables
`New-NetFirewallRule -DisplayName "BLK 198.51.100.7" -Direction Outbound -RemoteAddress 198.51.100.7 -Action Block`	Windows (repeat with Inbound)

ROSETTA Hit counters, sessions, capture

COMMAND	PLATFORM
Rule hit counters	
`show firewall name WAN_IN statistics`	Vyatta
`show access-lists WAN_IN`	Cisco IOS (matches per line)
`show access-list OUTSIDE_IN`	Cisco ASA (hitcnt)
`show security policies hit-count`	Junos SRX
`show rule-hit-count vsys vsys-name vsys1 rule-base security rules all`	PAN-OS
`/ip firewall filter print stats`	MikroTik
`pfctl -vvsr  iptables -L -n -v`	pfSense / Linux
See / kill sessions for an IP	
`show conntrack table ipv4 | match IP` `sudo conntrack -D -s IP`	Vyatta / Linux
`show conn address IP  clear conn address IP`	Cisco ASA
`show security flow session source-prefix IP  clear security flow session source-prefix IP`	Junos SRX
`diagnose sys session filter src IP; diagnose sys session list; diagnose sys session clear`	FortiOS
`show session all filter source IP` `clear session all filter source IP`	PAN-OS
`/ip firewall connection remove [find src-address~"IP"]`	MikroTik
`pfctl -ss | grep IP  pfctl -k IP`	pfSense
Packet capture on the box	
`sudo tcpdump -ni eth0 host IP -w /tmp/c.pcap`	Vyatta, pfSense, Linux
`monitor capture CAP interface Gi1 both match ipv4 host IP any; monitor capture CAP start | stop; monitor capture CAP export flash:c.pcap`	Cisco IOS-XE
`capture CAP interface outside match ip host IP any; copy /pcap capture:CAP disk0:/c.pcap`	Cisco ASA
`diagnose sniffer packet wan1 'host IP' 6 0 l`	FortiOS (text; convert to pcap offline)
`/tool sniffer quick ip-address=IP`	MikroTik
`pktmon filter add -i IP; pktmon start --capture; pktmon stop; pktmon etl2pcap PktMon.etl`	Windows 10 2004+

CISCO IOS / IOS-XE Named ACLs: edit in place

```
show ip interface brief
show running-config | include access-group
show access-lists WAN_IN                          ! entries with
   sequence numbers + matches
copy running-config flash:pre-CR123.cfg
```

CISCO IOS / IOS-XE Named ACLs: edit in place (cont.)

```
configure terminal
ip access-list extended WAN_IN
 remark CR123 block C2 198.51.100.7
 5 deny ip host 198.51.100.7 any log            ! insert by sequence
   number
 15 permit tcp any host 203.0.113.10 eq 443
 no 30                                          ! remove a single
   entry
 1000 deny ip any any log                       ! visible explicit
   deny
exit
ip access-list resequence WAN_IN 10 10          ! renumber
   10,20,30...
interface GigabitEthernet0/0
 ip access-group WAN_IN in
end
show ip interface Gi0/0 | include access list
write memory
```

SYNTAX	MEANING
`permit|deny PROTO SRC [ports] DST [ports] [established] [log]`	extended entry layout
`host 10.0.0.5  any  10.0.0.0 0.0.0.255`	one host / everything / wildcard mask (/24)
`eq 443  range 8000 8100  gt 1023 neq 23`	port operators
`permit tcp any any established`	TCP replies only (stateless ACK/RST check)
`object-group network BLOCKLIST / host 198.51.100.7`	group, used as `deny ip object-group BLOCKLIST any`
`ip access-list standard MGMT / permit 10.0.99.0 0.0.0.255` `line vty 0 15 / access-class MGMT in / transport input ssh`	lock down SSH to the router
`ip nat inside source static tcp 10.0.0.10 443 interface Gi0/0 443`	port forward
`ip nat inside source list NAT_ACL interface Gi0/0 overload`	PAT / masquerade

Gotchas: `no access-list 101` deletes the *whole* numbered ACL, so use named ACLs and edit by sequence. Applying an ACL that does not exist yet permits everything. The implicit final `deny ip any any` has no counter. One ACL per interface, per direction, per protocol.

CISCO IOS / IOS-XE Verify, logs & DFIR

```
show version ; show clock detail ; show users
show logging | include IPACCESSLOG                  ! %SEC-6-
   IPACCESSLOGP = ACL log hits
show archive log config all                         ! who changed
   what (if enabled below)
show running-config | include username|snmp-server community|event
   manager|kron
show event manager policy registered                ! EEM =
   persistence / triggers
show control-plane host open-ports ; show tcp brief all
show ip nat translations ; show ip route ; show arp
show processes cpu sorted ; dir flash: ; verify /md5 flash:
   <image.bin>
show tech-support | redirect flash:tech-CR123.txt
! enable config change logging + archive (needed for revert timer)
archive
 path flash:archive-cfg
 write-memory
 log config
  logging enable
  notify syslog contenttype plaintext
  hidekeys
```

CISCO ASA (AND FTD CLI) ACLs, objects, packet-tracer

```
show running-config access-group
show running-config access-list OUTSIDE_IN
show access-list OUTSIDE_IN                         ! line numbers
   + hitcnt
copy running-config disk0:/pre-CR123.cfg
configure terminal
object network WEB1
 host 10.0.0.10
object-group network BLOCKLIST
 network-object host 198.51.100.7
access-list OUTSIDE_IN line 1 extended deny ip object-group
   BLOCKLIST any log
access-list OUTSIDE_IN line 2 extended permit tcp any object WEB1 eq
   https
access-list OUTSIDE_IN remark CR123 2026-10-21
no access-list OUTSIDE_IN extended permit tcp any host 10.0.0.99 eq
   3389   ! remove an entry
access-group OUTSIDE_IN in interface outside
```

CISCO ASA (AND FTD CLI) ACLs, objects, packet-tracer (cont.)

```
end
write memory
! simulate a packet through every phase (ACL, NAT, route) — no
   traffic sent
packet-tracer input outside tcp 198.51.100.7 40000 10.0.0.10 443
   detailed
! incident tools
shun 198.51.100.7 ; show shun ; no shun 198.51.100.7
show conn address 198.51.100.7 ; clear conn address 198.51.100.7
show logging | include 106023                       ! denied by
   access-group
show xlate ; show nat detail
```

- 8.3+ ACLs use the **real** (internal) IP, not the mapped/public one.
- Without an ACL, traffic flows from higher to lower `security-level`. Once an ACL is bound, only the ACL decides.
- FTD is managed in FMC/FDM: change rules there. `packet-tracer` and `capture` also work in the FTD CLI.

JUNIPER JUNOS (SRX) Security policies & commit

```
configure private                                   # own candidate
   copy
show security policies | display set
set security address-book global address BAD7 198.51.100.7/32
set security address-book global address-set BLOCKLIST address BAD7
set security policies from-zone untrust to-zone trust policy BLOCK-
   BAD match source-address BLOCKLIST
set security policies from-zone untrust to-zone trust policy BLOCK-
   BAD match destination-address any
set security policies from-zone untrust to-zone trust policy BLOCK-
   BAD match application any
set security policies from-zone untrust to-zone trust policy BLOCK-
   BAD then deny
set security policies from-zone untrust to-zone trust policy BLOCK-
   BAD then log session-init
insert security policies from-zone untrust to-zone trust policy
   BLOCK-BAD before policy ALLOW-WEB
show | compare
commit check
commit confirmed 10 comment "CR123"
commit                                              # confirm
   within 10 min
rollback 1 ; commit                                 # undo the last
   commit
# op mode
show security match-policies from-zone untrust to-zone trust source-
   ip 198.51.100.7 destination-ip 10.0.0.10 source-port 40000
   destination-port 443 protocol tcp
show security policies hit-count
show security flow session source-prefix 198.51.100.7
show system commit ; show configuration | compare rollback 3
show system users ; show log messages | last 200
show configuration event-options ; show configuration system scripts
   # persistence
```

Stateless filters (routers, EX, protect the RE): `set firewall family inet filter PROTECT-RE term ...`, then `set interfaces lo0 unit 0 family inet filter input PROTECT-RE`. Counters: `show firewall`.

FORTINET FORTIGATE (FORTIOS) Policies, VIPs, flow debug

```
show firewall policy                                # order = list
   position, not ID
config firewall address
    edit "BAD7"
        set subnet 198.51.100.7 255.255.255.255
    next
end
config firewall addrgrp
    edit "BLOCKLIST"
        append member "BAD7"
    next
end
config firewall policy
    edit 0                                          # 0 = next free
   ID
        set name "CR123-block-bad"
        set srcintf "wan1"
        set dstintf "any"
        set srcaddr "BLOCKLIST"
        set dstaddr "all"
        set action deny
        set schedule "always"
        set service "ALL"
        set logtraffic all
    next
    move 42 before 3                                # reorder by ID
end
config firewall vip                                 # port forward
    edit "WEB1-443"
        set extintf "wan1"
```

FORTINET FORTIGATE (FORTIOS) Policies, VIPs, flow debug (cont.)

```
        set mappedip "10.0.0.10"
        set portforward enable
        set extport 443
        set mappedport 443
    next
end
# why was it allowed / denied?
diagnose debug flow filter addr 198.51.100.7
diagnose debug flow show function-name enable
diagnose debug enable
diagnose debug flow trace start 20
diagnose debug disable
# sessions + logs
diagnose sys session filter src 198.51.100.7
diagnose sys session list ; diagnose sys session clear
execute log filter category traffic ; execute log display
get system status ; show system admin ; show system automation-
   stitch   # admins + automation persistence
```

PALO ALTO PAN-OS Security rules from the CLI

```
set cli config-output-format set                    # show config
   as set-commands
configure
save config to pre-CR123.xml
set address BAD7 ip-netmask 198.51.100.7/32
set address-group BLOCKLIST static [ BAD7 ]
set rulebase security rules CR123-BLOCK from untrust to any source
   BLOCKLIST destination any application any service any action deny
   log-end yes
move rulebase security rules CR123-BLOCK top
show rulebase security
validate full
commit description "CR123"
revert config                                       # discard
   uncommitted candidate changes
load config from pre-CR123.xml ; commit             # back out
exit
# op mode
test security-policy-match from untrust to trust source 198.51.100.7
   destination 10.0.0.10 protocol 6 destination-port 443
show rule-hit-count vsys vsys-name vsys1 rule-base security rules
   all
show session all filter source 198.51.100.7 ; clear session all
   filter source 198.51.100.7
show admins all ; show jobs all
show log config direction equal backward            # config audit:
   who changed what
show log system direction equal backward
```

- Default rules: *intrazone-default* allows and *interzone-default* denies, and neither logs. Override them to enable logging.
- Rules use the pre-NAT IP with the post-NAT zone. For blocklists, an External Dynamic List (EDL) or a Dynamic Address Group needs no commit per IOC.

MIKROTIK ROUTEROS Filter rules & IR checks

```
# Ctrl+X = Safe Mode: changes revert if the session drops. Ctrl+X
   again keeps them.
/export file=pre-CR123
/ip firewall filter print stats
/ip firewall address-list add list=BLOCKLIST address=198.51.100.7
   comment="CR123"
/ip firewall filter add chain=input connection-
   state=established,related action=accept place-before=0
/ip firewall filter add chain=forward src-address-list=BLOCKLIST
   action=drop log=yes log-prefix="BLK" place-before=0
/ip firewall filter add chain=forward dst-address-list=BLOCKLIST
   action=drop log=yes log-prefix="BLK"
/ip firewall filter add chain=input in-interface-list=WAN
   action=drop comment="final drop"
/ip firewall filter move 7 destination=1            # reorder
/ip firewall filter disable 7 ; /ip firewall filter remove 7
/ip firewall nat add chain=dstnat in-interface=ether1 protocol=tcp
   dst-port=443 action=dst-nat to-addresses=10.0.0.10 to-ports=443
/ip firewall connection print where src-address~"198.51.100.7"
/log print where message~"BLK"
# DFIR: common implant / botnet persistence spots
/user print ; /user active print ; /ip service print
/system scheduler print ; /system script print ; /file print
/ip socks print ; /ip proxy print ; /ip dns static print
/ip firewall nat print ; /interface print ; /system package print
```

Chains: input = to the router, forward = through it, output = from it. Chains **accept by default**, so they need a final drop. Chains are evaluated top-down.

PFSENSE / OPNSENSE GUI rules & pf shell

- **Firewall › Rules › <interface>**: rules match traffic *entering* that interface, top-down, first match. Floating rules are evaluated first. Click **Apply Changes**.
- **Firewall › Aliases**: IP/port lists (URL tables for blocklists). Edit the alias rather than the rules.

PFSENSE / OPNSENSE GUI rules & pf shell (cont.)

- LAN has an *anti-lockout* rule. WAN denies everything unsolicited by default. Port forwards create a linked pass rule that matches the internal IP.
- Config history: Diagnostics › Backup & Restore › Config History (diff / revert). Console menu option **15** restores a recent config (pfSense).

```
pfctl -sr ; pfctl -vvsr                             # rules (+ counters,
   rule numbers)
pfctl -ss | grep 198.51.100.7 ; pfctl -k 198.51.100.7   # states /
   kill
pfctl -t BLOCKLIST -T add 198.51.100.7              # runtime only
pfctl -t BLOCKLIST -T show ; pfctl -si
tail -f /var/log/filter.log                         # Status › System
   Logs › Firewall
pfctl -d                                            # EMERGENCY:
   firewall + NAT off; pfctl -e to re-enable
# config: pfSense /cf/conf/config.xml (+ backup/) ; OPNsense
   /conf/config.xml (+ backup/)
```

LINUX iptables / ipset / nftables / ufw / firewalld

```
iptables -L -n -v --line-numbers ; iptables -t nat -L -n -v
iptables-save > /root/pre-CR123.rules ; iptables-restore <
   /root/pre-CR123.rules
iptables -I INPUT 1 -s 198.51.100.7 -j LOG --log-prefix "BLK "
iptables -I INPUT 2 -s 198.51.100.7 -j DROP
iptables -D INPUT 3                                  # delete by line
   number
ipset create BLOCKLIST hash:ip ; ipset add BLOCKLIST 198.51.100.7
iptables -I FORWARD -m set --match-set BLOCKLIST src -j DROP
# nftables (table/chain names vary: nft list ruleset first)
{ echo "flush ruleset"; nft list ruleset; } > /root/pre-CR123.nft
nft -a list chain inet filter input                  # -a shows rule
   handles
nft insert rule inet filter input ip saddr 198.51.100.7 counter drop
nft delete rule inet filter input handle 12
nft add set inet filter blocklist '{ type ipv4_addr; }'
nft add element inet filter blocklist '{ 198.51.100.7 }'
# ufw / firewalld front-ends
ufw status numbered ; ufw insert 1 deny from 198.51.100.7 ; ufw
   delete 3
firewall-cmd --list-all
firewall-cmd --add-rich-rule='rule family=ipv4 source
   address=198.51.100.7 drop'   # runtime
firewall-cmd --runtime-to-permanent
# sessions
conntrack -L -s 198.51.100.7 ; conntrack -D -s 198.51.100.7
```

WINDOWS DEFENDER FIREWALL PowerShell / netsh & host isolation

```
Get-NetFirewallProfile | ft
   Name,Enabled,DefaultInboundAction,DefaultOutboundAction,LogBlocke
   d
netsh advfirewall export C:\IR\pre-CR123.wfw          # import to
   restore
New-NetFirewallRule -DisplayName "CR123 BLK in" -Direction Inbound -
   RemoteAddress 198.51.100.7 -Action Block
New-NetFirewallRule -DisplayName "CR123 BLK out" -Direction Outbound
   -RemoteAddress 198.51.100.7 -Action Block
Get-NetFirewallRule -DisplayName "CR123*" | Remove-NetFirewallRule
Get-NetFirewallRule -Enabled True -Direction Inbound -Action Allow |
  Get-NetFirewallPortFilter | ? LocalPort -ne Any       # open
   inbound ports
Set-NetFirewallProfile -All -LogBlocked True -LogAllowed True -
   LogMaxSizeKilobytes 32767
# log: %SystemRoot%\System32\LogFiles\Firewall\pfirewall.log
# isolate: block everything except the IR box 10.0.0.5 (block beats
   allow)
$r = "0.0.0.0-10.0.0.4","10.0.0.6-255.255.255.255"
New-NetFirewallRule -DisplayName "IR-ISOLATE out" -Direction
   Outbound -RemoteAddress $r -Action Block
New-NetFirewallRule -DisplayName "IR-ISOLATE in"  -Direction Inbound
   -RemoteAddress $r -Action Block
Get-NetFirewallRule -DisplayName "IR-ISOLATE*" | Remove-
   NetFirewallRule   # release
```

Evidence: Security 4946 / 4947 / 4948 (rule added / changed / deleted), 4950 (setting changed), 5156 / 5157 (WFP allowed / blocked; need auditing on). Firewall log: Microsoft-Windows-Windows Firewall With Advanced Security/Firewall 2004 / 2005 / 2006 (rule added / changed / deleted).

REFERENCE **CIDR, netmask, Cisco wildcard**

CIDR	NETMASK	WILDCARD	CIDR	NETMASK	WILDCARD
/32	255.255.255.255	0.0.0.0	/24	255.255.255.0	0.0.0.255
/31	255.255.255.254	0.0.0.1	/23	255.255.254.0	0.0.1.255
/30	255.255.255.252	0.0.0.3	/22	255.255.252.0	0.0.3.255
/29	255.255.255.248	0.0.0.7	/21	255.255.248.0	0.0.7.255
/28	255.255.255.240	0.0.0.15	/20	255.255.240.0	0.0.15.255
/27	255.255.255.224	0.0.0.31	/16	255.255.0.0	0.0.255.255
/26	255.255.255.192	0.0.0.63	/12	255.240.0.0	0.15.255.255
/25	255.255.255.128	0.0.0.127	/8	255.0.0.0	0.255.255.255

Wildcard = 255.255.255.255 minus the netmask. RFC1918: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. Also bogons: 100.64.0.0/10 (CGNAT), 169.254.0.0/16 (link-local), 127.0.0.0/8, 0.0.0.0/8, 224.0.0.0/4 (multicast), 240.0.0.0/4. Documentation ranges used on this sheet: 192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24.